

Machine Learning for Human–Machine Systems With Advanced Persistent Threats

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Abstract—This article conducts a thorough exploration of the implications of machine learning (ML) in conjunction with human–machine systems within the military domain. It scrutinizes the strategic development efforts of ML by pertinent institutions, particularly in the context of military applications and the domain of advanced persistent threats. Prominent nations have delineated a technical trajectory for the integration of ML into their military frameworks. To bolster the structure and efficacy of their various military branches and units, there has been a concentrated deployment of numerous ML research endeavors. These initiatives encompass the study of sophisticated ML algorithms and the acceleration of artificial intelligence technology adaptation for intelligence processing, autonomous platforms, command and control infrastructures, and weapons systems. Forces across the globe are actively embedding ML technologies into a range of platforms—terrestrial, naval, aerial, space-faring, and cybernetic. This integration spans weaponry, networks, cognitive operations, and additional systems. Furthermore, this article reviews the incorporation within the sphere of military human–machine interaction in the Russia–Ukraine conflict. In this war, cyber human–machine interaction has become a pivotal arena of contention between Russia and Ukraine, with key levers that influence the conflict’s course. In addition, the article examines the adoption of ML in prospective military functions such as, operations, intelligence gathering, networking, logistics, identification protocols, healthcare, data analysis trends, and other critical areas marked by current developments

and trajectories. It also proffers a series of recommendations for the future integration of ML to inform strategic direction and research.

Index Terms—Artificial intelligence (AI) strategy, future trend, machine learning (ML), research project.

I. INTRODUCTION

IN THE domain of machine learning (ML) research and application, it is imperative for institutions [1] associated with global military power to monitor and assess advancements in cutting-edge ML technologies. Such entities must bolster strategic advisory services, technological inquiry [2], and concerted efforts in military deployment [3], [4]. Defense departments across nations are actively cultivating ML research initiatives. Moreover, ML substantially augments the efficiency of various operational facets, including network-centric warfare, intelligence, surveillance, logistics, medical services, simulations, and situational data analytics [5]. The prospective evolution of military technology through ML integration holds profound strategic importance and presents a rich vein of research potential [6].

A. Assessing ML Advancements in Global Military Power

Our study initiated a comprehensive literature review, incorporating engineering case studies to scrutinize the prevailing research findings and the actual combat deployment advancements within the context. This investigation entailed a systematic tracking and evaluative analysis of global military ML technological advancements. We compiled a detailed account of the current state of global military ML technology, dissected the principal technological trajectories and trends, and offered an in-depth examination of emblematic military applications of global ML technology, alongside their developmental prospects. The aim of this article is to elucidate the implementation of ML in military human–machine systems and allied domains, with the intent to either spark or amplify interest among readers. It also delves into artificial intelligence (AI) military research pertaining to human–machine systems. Notably, it presents an overview of the evolution of human–machine interaction as evidenced in the Russia–Ukraine conflict. The deployment of next-generation network technologies in military human–machine interaction contexts is examined. Ultimately, we synthesize the future outlook for each application scenario.

B. Military Background

In recent years, AI has emerged as a pivotal force with the potential to revolutionize future warfare. Nations worldwide are

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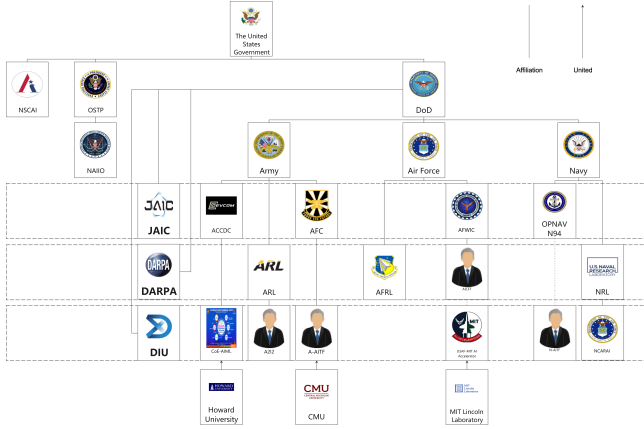


Fig. 1. U.S. ML force.

ving for supremacy in AI technology, recognizing its strategic importance in reshaping military power [7]. Furthermore, countries have acknowledged the integral role of AI in bolstering their national and military strategic initiatives. These nations are intensifying their investments in innovation and infrastructure to augment their AI capabilities and are engaging in both theoretical and applied AI research to amplify their armed forces' operational effectiveness [8]. The evolution of the military's capabilities across various domains has traditionally adhered to a methodical scientific progression. This includes strategizing at the highest level, establishing core forces, supporting capabilities or resources, and executing combat operations. Presently, the countries are focused on refining its military AI strategy, cultivating an AI-savvy military workforce, fostering military-civil integration of AI technologies, and actively incorporating AI methodologies into diverse military spheres.

C. Current Status of Military Research on ML

Taking some countries as a case study (Fig. 1), there has been a significant integration of AI within its military technology framework in recent years. The Joint Artificial Intelligence Center (JAIC) has been instrumental in propelling the advancement of next-generation AI technologies to enhance combat capabilities. Concurrently, the Defense Advanced Research Projects Agency (DARPA) has embarked on foundational research in AI sciences, setting the stage for future breakthroughs.

Globally, military powerhouses have been meticulously monitoring and assessing the strategic importance of ML technology, intensifying their efforts in consultation, research, and practical application. This article delves into the evolution of ML evaluation within the military sector. A comprehensive set of initiatives has been adopted to advance AI deployment in defense contexts, including the establishment of overarching national strategies and the release of numerous strategic plans for intelligence development. A variety of military intelligence research endeavors have been undertaken, with a focus on achieving breakthroughs in core technologies, thereby, providing a robust foundation for the growth of military intelligence capabilities. Specifically, the United States Department of Defense (DoD) has expedited the application of AI in military operations, actively

enhancing operational efficiency by leveraging AI to augment, rather than supplant, military personnel within information systems. The foundation of the JAIC stands as a testament to these efforts, orchestrating and fostering AI-related military research. In the current phase of ML's military application, particularly in the comprehensive research on autonomous systems, ML has demonstrated considerable superiority [9]. Moreover, the fusion of human and ML is anticipated to offer even more substantial benefits in the future, claiming an increasing share in defense applications. This convergence is likely to spur rapid technological innovation in the military sphere, striving for an advanced level of integration between human capabilities and AI. Notably, the interplay between human cognition and AI's computational prowess is advancing AI applications and poised to have far-reaching implications in defense sectors. The prospect of sophisticated human-AI integration holds vast potential for future military endeavors.

II. METHODOLOGY

The methodology integrates a comprehensive approach to data collection and analysis. The methodology is structured as follows.

A. Theoretical Foundation

It is built upon a solid theoretical base and incorporates the latest findings related to advanced persistent threats (APTs) in military systems. This review informs our selection of features and ML techniques pertinent to APT detection and response.

B. Data Collection and Preprocessing

Our data collection is formalized as $D_{\text{collection}} = \bigcup_{i=1}^N S_i$, $D_{\text{preprocessed}} = f_{\text{preprocess}}(D_{\text{collection}})$, where $D_{\text{collection}}$ is the union of datasets from N sources S_i , and $f_{\text{preprocess}}$ represents the preprocessing operations, including data cleansing, normalization, and initial feature selection, all of which are informed by the insights gained from the literature review.

C. Feature Engineering and Selection

$$F = f_{\text{feature}}(D_{\text{preprocessed}}, L_{\text{insights}})$$

where F represents the engineered feature set, with f_{feature} being the feature engineering function that leverages both the preprocessed data and insights L_{insights} from the literature to identify features that are most indicative of APT activities.

D. Model Construction and Optimization

$$M = f_{\text{model}}(F, \theta, L_{\text{insights}})$$

where M denotes the machine learning models constructed for APT detection. The function f_{model} utilizes the selected features F , model parameters θ , and insights from literature L_{insights} to train and optimize models that are robust against the sophisticated nature of APTs.

E. Model Evaluation and Validation

$$E = f_{\text{evaluate}}(M, D_{\text{test}}, L_{\text{metrics}})$$

where E signifies the evaluation results of the models, with f_{evaluate} incorporating a set of evaluation metrics L_{metrics} derived from literature, such as, precision, recall, and F1-score. These metrics are applied to a test dataset D_{test} to validate model performance and ensure reliability in operational settings.

III. ML PROJECT TRACKING IN MILITARY FIELDS

In comparison to other nations, the military ML initiatives stand out in both the volume of equipment research and the sophistication of their technological research, deployment, and development efforts. This article aims to meticulously track and analyze the current ML projects within the US military. It encapsulates the strategic initiatives and consultations by various U.S. defense entities and the Russian government regarding the integration and evaluation of ML in military operations from 2018 to 2020. The U.S. Land (NDRI) in 2019 focused on assessing AI technology for combat applications to strengthen the DoD AI capabilities [10]. In 2018, the U.S. DoD outlined a technical roadmap for ML in cyber operations [11], emphasizing defensive cybersecurity as a foundation for AI utilization. The U.S. Congressional Research Service in the same year raised concerns about adversaries using Lethal Autonomous Weapon Systems [12] and advocated for a clear strategy on autonomous systems and AI [13]. The U.S. JAIC, also in 2018, aimed to promote ML to expedite critical operational tasks and establish a shared infrastructure for ML in the DoD. The U.S. Defense Innovation Unit sought to leverage commercial tech innovation for defense purposes, focusing on computer vision, large database analysis, and strategic reasoning [14]. Concurrently, in 2018, the Russian Government Military Industry Committee laid out plans for military automation, with an aim to achieve 30% automation in military technology by 2025. The Ministry of Digital Development of Russia in 2020 announced funding for AI projects within its digital transformation agenda, detailing projects across four ministries and three government departments. Lastly, the Russian Ministry of Defense's 2020 meeting underscored the pivotal role of AI in weapon control for future warfare.

Table I details service cases of ML applications across various military branches from army to cyber forces, illustrating the diverse technological advancements and projects undertaken. The Army has been involved in autonomous networking, fog computing, and applied ML systems to enhance data prioritization and reduce cognitive burdens. The Navy has embarked on projects like TOPGUN for improved target detection and ship identification, while the Air Force has pursued initiatives like Skyborg and the sixth-generation fighter employing advanced computational intelligence and ML technologies. The Marines have integrated deep reinforcement learning into weapon systems, and the Space Army has focused on space-based surveillance and real-time image analysis using ML algorithms. Cyber Forces have developed automated cyber defense tools like CHASE and deepArmor, as well as social media manipulation defenses using Big Data analytics.

The military has strategically initiated a multitude of military research projects, delving into intelligent ML algorithms and fostering the integration of AI technology across various domains, including intelligence processing, autonomous combat platforms, command and control systems, weaponry, and combat strategies (Table I). The Army has made strides in independent network operations and distributed processing within heterogeneous environments, applying ML to system research. Through innovative approaches such as, few-shot learning, autonomous networks, fog edge computing, and cognitive science, there is an ongoing endeavor to marry the latest ML research with the unique tactical requirements of the US military. The Navy has implemented AI in naval information systems to enhance capabilities in electronic warfare, focusing on detection, identification, sensing, and defense against emergent threats. The Air Force's Quantum Project, along with initiatives in autonomous swarm networks, ML 'wingmen', and development of sixth-generation aircraft, has significantly bolstered its combat prowess, showcasing an edge in functional infrastructure and R&D deployment.

The Marine Corps has explored the potential of reinforcement learning in collaborative information warfare, integrating ML-assisted cognitive decision-making within its operational framework to refine assessments of its weapons systems [15]. The Space Force has engaged in space-based data analytics and processing, leveraging ML on supercomputing and cloud platforms to augment its capacity for handling vast quantities of real-time data. Notably, the Cyber Command has formulated comprehensive plans addressing multiple dimensions such as, cyber threat defense, IoT security in military contexts, ML-based behavior detection, social network data analysis, and cyber electronic warfare, significantly enhancing capabilities in network offense and defense, social network behavior manipulation, and electronic warfare responsiveness (Table I). Current ML applications are predominantly utilized in "back office" processes rather than direct frontline engagement.

Table II presents preresearch projects on AI with an emphasis on human-machine integration. It includes projects like Human-Machine Communication (CWC), which aims at advancing machines' understanding of language and gestures, TOPGUN for naval information processing, and Complex Hybrid System (CHS) for developing theories and algorithms for complex system analysis. The ML Wingman Project (Skyborg) seeks to create an active "Machine Brain" for human-machine symbiosis, while the TACE system connects human intelligence with machine autonomy for aircraft piloting. Lastly, Dynamic Airspace Control (DAC) explores methods for managing airspace without reliance on high-power radar systems.

In recent times, the Human-Machine Systems has actively engaged in AI research aimed at revolutionizing combat theory and advancing the development of intelligent equipment. Table II denotes the AI preresearch projects that have been undertaken in recent years. By harnessing AI for fusion empowerment, knowledge reasoning, exploration, and interpretation, these algorithms have become self-sufficient. This reflects the evolving defense-related preresearch trends and characteristics from a human-machine integration perspective that is adaptive to various platforms.

TABLE I
SERVICE CASES OF MILITARY ML APPLICATIONS

Service arms	Project	Technology
Army	Information collection for autonomous network projects	Automatic network decision security, Small sample learning, Adaptive and autonomous defense against cyber attacks
Army	DPHTE	Fog computing platform, Integrate smart edge device data
Army	Applied ML system	Reduce cognitive burden, Small sample target detection, Prioritize data, Automatically filter out unimportant signals
Army	Russian Ministry of Defense tests AI targets for military training	Modern aiming complex Autonomous targets with artificial intelligence algorithms, Tracking system can simulate allied or enemy forces
Navy	Global unmanned naval information processing project (TOPGUN)	Detect farther targets, Identify more types of ships, Water display performance
Air Force	Air Force "Quantum Project "	Computational Intelligence, Autonomous reasoning and decision making, Relevant social impact
Air Force	DPAA(defense POW/MIA accounting agency)	Research on autonomous cluster elastic network, The distributed phased array antenna system of the project
Air Force	Skyborg	Independent, can attack and open the UAV
Air Force	Sixth-generation fighter	Computer miniaturization, ML technology, Collection, compilation and processing ISR data
Air Force	U.S. Air Force industry-university integration innovation agency	Real-time analysis, Deep reinforcement learning intelligence, AI virtual base, Create comprehensive global coverage of surveillance and reconnaissance (ISR) system system, Identify cutting-edge commercial satellite payload concepts, designs and prototypes
Air Force	Russia's large attack drone	Technical vision, Autonomous driving capabilities, Potential manned and unmanned collaboration
Marines	Deep reinforcement learning technology is integrated into USMC's weapon system	Deep reinforcement learning technology, Integrated into USMC, Weapon system
Marines	Sea surface automatic analysis system	Satellite and terrestrial information systems, Radar station data and internal channels to transmit information, Receive data from over-the-horizon radar
Space Army	Onboard data processing and interpretation	Space-based surveillance system, All day, Continuous work
Space Army	Real-time high-efficiency analysis of space remote sensing images	Cloud system and software for supercomputers, ML and artificial intelligence algorithms, Raw data from space observation sensors, Analysis of current affairs images
Cyber Forces	CHASE (cyber hunting at scale)	Develop, demonstrate and evaluate new automated cyber defense tools [16], ML and cyber attack modeling, Automatic detection and defense against currently undetected advanced threats
Cyber Forces	Cyber Forces Military IoT cyber defense platform deepArmor	ML to identify and analyze unknown files, Detect military IoT network of malicious threats
Cyber Forces	Einstein project	Behavior detection capabilities based on ML
Cyber Forces	Social media manipulate	Processing and analysis of social media big data, Natural language processing technology, Image and emotion recognition technology, Improve the scale and speed of collecting and processing social media data
Cyber Forces	Cyberspace electronic warfare	EWPMPT, ML upgrade and improvement

IV. ML METHODOLOGY FOR MILITARY AND APT WITH HUMAN-MACHINE SYSTEMS

The integration of human-machine systems within military operations (Table III) has seen a swift evolution, propelled by cutting-edge research [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30]. The defense industry

necessitates a novel paradigm for the natural interaction between human operators and autonomous aerial vehicles (UAVs) to strike an optimal balance between drone control and the cognitive load on operators. The optimization of human-machine co-operation is particularly applicable in scenarios such as weapon-to-target assignments, where it must account for complex multiagent interactions, intricate information landscapes, and

TABLE II
AI PRERESEARCH PROJECTS WITH HUMAN–MACHINE INTEGRATION

Project	Content
Human–Machine Communication (CWC)	Enable machines to understand language, vision, gestures and dialogue management, and advance the cutting-edge cognitive capabilities of HMI
Global Unmanned Naval Information Processing Project (TOPGUN)	Support independent development of TOPGUN, expand detection and classification capabilities, and use efficient deep learning algorithms to detect farther targets
Complex Hybrid System (CHS)	Develop basic quantitative theories and algorithms for complex system analysis and design, and verify and evaluate innovative test functions for these theoretical values through cross-domain experiments
ML Wingman Project (Skyborg)	Active "Machine Brain", Human-Machine Symbiosis (HMS) Project-Successful Team Artificial Social Intelligence (ASIST), demonstrates the thinking theory of machines and the ability to interact with team members by sharing mental models, TACE system, Connects human intelligence machines and aircraft autopilot.
Human–Machine Symbiosis Project-Successful Team Artificial Social Intelligence (ASIST)	Demonstrate the thinking theory of machines and the ability to interact with team members by sharing mental models
TACE (Testing of Autonomy in Complex Environments) system	Connect the autonomous system/AI computer and the aircraft autopilot. The TACE system is designed to monitor the commands sent from the autonomous system to the aircraft autopilot and provide information on the status of the aircraft
Dynamic Airspace Control (DAC)	Research on methods for monitoring and managing local airspace without using high-power radar

TABLE III
REVIEW OF ML FOR MILITARY WITH HUMAN–MACHINE SYSTEMS

Field	Main Author	View	Reference
Human-AI symbiotic partnership	Hou, M.	Frontiers of Brain-Inspired Autonomous Systems. Defence R&D drives the innovations.	[31]
Human–Machine collaborative optimization via apprenticeship scheduling	Gombolay, M.	Learning strategies learned from manual scheduling demos through apprenticeships can greatly improve the efficiency of progress optimization. This article applies this human-machine collaborative optimization technique to variations of weapon-to-target distribution issues.	[17]
DAIS-ITA Scenario	White, G.	This article lists military scenarios and operational examples within a future time frame to capture the expected battlefield environment and key challenges ahead. Key considerations involve complex multi-participant situations, high-complexity information, high-tempo processing, all in a mixed human-machineteam.	[18]
New generation of human machine interfaces for controlling UAV through depth based gesture recognition	Montecon, T.	A new form of natural interaction between human operators and UAVs is a requirement of the militaryindustry to achieve a better balance between drone control and the burden on human operators. In this paper, a new gesture recognition system based on depth image is proposed for the control of drones by human-machine interface (HMI).	[19]
Estimating cognitive load from speech gathered in a complex real-life training exercise	Vukovic, M.	Estimating cognitive load from speech is a type of speaker feature that provides insight into the state of human beings in complex, highly dynamic human-machine collaboration scenarios and is used to adapt interactions with users to their current cognitive state. Cognitive load estimation from speech experiments is typically performed on speech collected in a lab environment. In contrast, the study was conducted on a real-life data set that was not created for cognitive load assessment. The speech was extracted from a recording of a military simulation exercise in which air combat managers communicated with the pilots of the simulated aircraft.	[20]
Collaborative human-machine analysis to disambiguate entities in unstructured text and structured datasets	Davenport, JH.	Intelligence analysts need rapid information fusion capabilities to develop and maintain accurate situational awareness and understanding of dynamic enemy threats in asymmetric military operations. These NLP(Natural Language Processing) entity analyses enhance military capabilities, as analysts can now use intelligence automatically extracted from unstructured text to create/update entity profiles, incorporating entity knowledge from structured and unstructured data sources.	[21]
Resilient design and operation of cyber physical systems with emphasis on unmanned autonomous systems	Vachtsevanos, G	Autonomous and autonomous systemstake centre stage in the research community, as self-driving cars are proliferating and their utility in all aspects of the military and civilian spheres is growing exponentially every year. Unfortunately, the development and application of elasticity and safety technologies in autonomous systems has not kept pace with their growth rates. Several factors hinder the deployment and adoption of autonomous systems. These include a lack of autonomy that can be relied upon, major challenges in the human-machine interface area Requires a lot of human intervention to manipulate and interpret sensor data, the need for emerging ML technologies, and most importantly, complex systems.	[22]
Collaborative autonomous sensing with Bayesians in the loop	Ahmed, N.	There is a strong push to develop intelligent unmanned autonomosity to complement human reasoning for applications such as wilderness search and rescue, militarysurveillance and robotic space exploration. Autonomous intelligence is not only a substitute for "dull, dirty and dangerous" work, it is also expected to cope with a range of uncertainties while working closely with humans in new situations. The robotic revolution firmly establishes Bayesian algorithms as the first to solve challenging perception, learning,and decision-making problems.	[23]
HMI of social networks	Long, C.	The scope of cyberspace weapons is expanding from the physical network domain to the cognitive information domain. Social public opinion weapons are mainly divided into six categories: Bot, Botnet, Troll, Manipulate real people and events, Cyborg, and Hacked or stolen. The combat practice shows that the effectiveness of social media even exceeds some traditional combat methods. The technical means, scale, scope, and precision of social media weapons have been continuously improved. It is gaining momentum to reshape the cyberspace security pattern of various countries fundamentally.	[32]

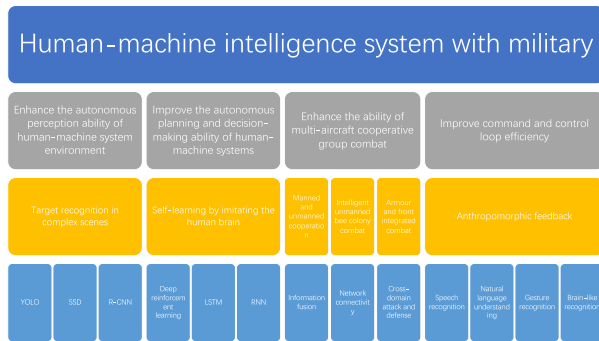


Fig. 2. Human-machine intelligence system with military.

rapid processing—all within a synergistic human-machine team setting.

The distinct benefits of human-machine collaboration in data processing highlight the value of a meticulously engineered “human in the loop” system, which can yield the most insightful revelations regarding patterns and workflows. One innovative approach for gauging the cognitive state of human operators in these dynamic collaborations is through speech analysis, which can infer cognitive load and adapt the interaction to the user’s current mental capacity.

A significant hurdle in human-machine interface development is the extensive human input required to manipulate and interpret sensor data. APTs present a novel challenge for both human and machine components. Crucially, the flexible design and operation of sophisticated systems must ensure they can navigate unstructured and unpredictable environments while maintaining safety, reliability, and serviceability.

This article encapsulates the latest progressions in the military domain concerning the integration of human and machine intelligence systems (Fig. 2).

The implementation of cutting-edge technologies such as, software-defined networking (SDN), information-centric networking (ICN), delay-tolerant networking (DTN), and the Internet of Battlefield Things (IoBT) is pivotal within the ambit of military human-machine interaction. These applications necessitate novel SDN frameworks capable of managing autonomous military equipment. SDN orchestrates the interaction patterns among heterogeneous nodes within the IoBT, while ICN is poised to fulfill the elevated Command and Control (C2) demands inherent in military mobile ad hoc networks [33].

V. HUMAN-MACHINE INTERACTION IN THE RUSSIA-UKRAINE WAR

The cyber domain of human-machine interaction has emerged as a critical arena in the ongoing conflict between Russia and Ukraine [34]. Advanced technologies such as satellite communications, autonomous systems, electronic countermeasures, social influence operations, and cyber warfare have become pivotal elements in the strategic calculus of this confrontation.

Our data indicates that in the first half of 2022, there were 22 disclosed incidents of APTs associated with the Russia-Ukraine conflict. These APTs were strategically engineered to infiltrate

key human-computer interaction interfaces, exfiltrating crucial interaction data between personnel and equipment to fulfill strategic objectives, thus, extending the battleground into the realm of information and human-machine interaction.

The conflict has redefined modern military engagement, highlighting the role of satellite communications, autonomous systems, electronic warfare, and cyber operations. It marks a departure from traditional warfare, which was primarily dominated by aircraft, tanks, warships, and missiles. In today’s context, marked by the continuous advent and mainstreaming of sophisticated information technologies, there is a clear shift towards a new paradigm of multidomain joint operations with human-machine intelligence serving as a pivotal link.

The strategic focus for future warfare will be to amplify research in key technological areas such as, ultralow orbit satellite systems, AI-driven countermeasures, directed energy weapons, and next-generation network security architectures for the battlefield. This focus is predicated on achieving dominance across electromagnetic, informational, and physical spheres. The evolution towards high-speed communication across domains and a digital military infrastructure is poised to redefine the digital battlespace, enabling a transparent operational environment, facilitating intelligent cross-domain collaboration of assets, and capturing the strategic high ground in modern warfare.

In conclusion, the deployment and efficacy of human-machine interaction and APT applications within military operations were comprehensively operationalized and empirically validated during the Russia-Ukraine War.

VI. FUTURE TRENDS, IMPLICATIONS, AND SUGGESTIONS

Armed forces across various nations are progressively embedding ML technologies into their diverse operational platforms, which include terrestrial, naval, aerial, space, and cybernetic domains. This scholarly review delineates the significant function of ML in pivotal military applications, projecting its impact on future military operations, intelligence gathering, network-centric warfare, logistical support, target identification, healthcare provision, data trend analysis, and other strategic areas of interest. The integration of ML is set to redefine these domains, offering unprecedented advancements in operational effectiveness, decision-making precision, and overall mission success in the landscape of modern warfare (Fig. 3).

A. Combat Systems

ML systems possess the potential to augment the efficacy of combat platforms [35], simultaneously enhancing the performance of combat systems and diminishing the reliance on human intervention. The command-and-control apparatus serves as the combat system’s central nervous system and is integral to triumph in warfare. Intelligent command and control methodologies can mitigate human limitations, enriching command efficacy and decision-making processes [36]. Through ML, comprehensive combat simulation analyses across various military branches and units can be conducted, facilitating the initiation of multiple sensor-simulation initiatives.

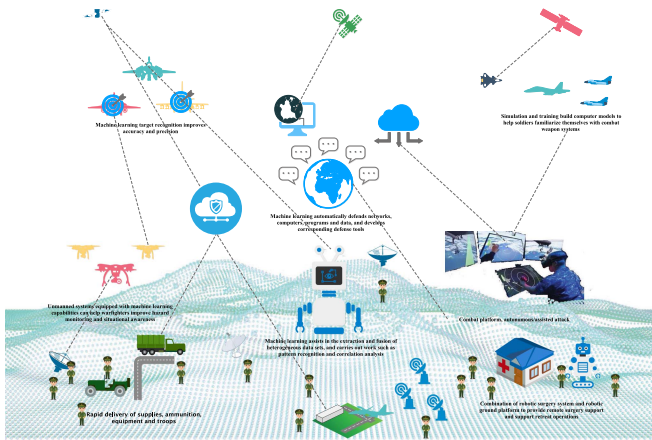


Fig. 3. Roles of ML in critical military applications.

B. Intelligent Weapons and Equipment

The strategic application of intelligent force can substantially boost both speed and precision, especially in scenarios characterized by antiaccess/area-denial constraints [24]. This application is predominantly utilized in an array of autonomous weaponry to forge a 3-D autonomous combat system. The symbiotic integration of human and machine epitomizes the quintessential synergy of mechanical accuracy and human ingenuity. Leveraging machine velocity and strength enables optimal human judgment, enhancing cognitive rapidity and precision. ML techniques further empower autonomous weaponry and the expeditious execution of coordinated assaults—intelligence in the realm of strike force applications. Progressively, automation, autonomy, and remote control functionalities of various autonomous systems and combat platforms will continue to achieve groundbreaking advancements.

C. Knowledge, Reasoning, and Cognition

Reconnaissance detection systems and intelligent sensing apparatuses are routinely employed in localized combat zones. The burgeoning volume of multisource, heterogeneous defense data poses a formidable challenge to conventional cognitive velocities and processing capacities [37]. Intelligent perception and pattern recognition technologies have attained or eclipsed human benchmarks; however, their cognitive competencies in areas such as logical reasoning and domain transfer remain underdeveloped. Research in interpretable AI and machine common-sense is pursued through the development of a multihypothesis “semantic engine,” which classifies and reinforces the learning of multimedia data events of significance, crafting computational models that learn from experiences and emulate core cognitive domains.

Moreover, physical AI’s automated extraction of scientific knowledge, well-founded AI language acquisition, and robust ML applications strive to quantify and synthesize diversity. Ideologies and psychologies are also potential targets for strategic influence through intelligent methodologies, adhering to ideological tenets to conduct management operations and “soft

strike” campaigns such as, cultural cold wars, political genetic engineering, and ideological subversion.

D. Algorithmic Warfare

Algorithms are deployed to address the offensive and defensive challenges of warfare, harnessing the immense potential of AI algorithms in situational awareness, intelligence analysis, command decision-making, and firepower coordination to secure victory. The knowledge capabilities intrinsic to ML systems are significantly influenced by the caliber of input data. The datasets employed for system training may be compromised or entail risks, with software susceptible to cyber assaults [38]. The cultivation of advanced hybrid intelligent algorithms is imperative to enhance system capabilities and ensure the secure operation of self-learning systems. ML can establish self-educating and adaptive technologies for life systems, addressing the challenges posed by machines adapting to novel data through continuous learning experiences. It can also construct test platforms for ML to counteract threat scenarios. The development of small-sample adversarial generation aligns more closely with the mutable and unpredictable nature of real-world environments. Subsequently, various forms of autonomous, human–machine coordinated, and cluster cooperative combat capabilities will experience significant enhancements, propelling a higher echelon of human–machine intelligent integrated warfare.

E. Human–Brain–Machine Synergy

Advancements in interface technologies, including neurosensory perception, motor skills, and peripheral and central nervous systems, are amplifying soldiers’ cognitive and decision-making faculties. By emulating the human cerebral cortex’s structure, significant strides in brain–machine interaction and control technologies have been achieved, markedly elevating the autonomous capabilities of autonomous military platforms. Human–machine communication has evolved to comprehend language, vision, gestures, and dialogue management, propelling the frontier of cognitive human–machine interaction, fostering the intricate development of foundational quantitative theories, algorithms for complex systems analysis and design, and facilitating cross-domain experiments to substantiate and appraise the theoretical merits of novel testing functionalities. A bespoke AI initiative has been inaugurated that amalgamates human expertise with data-derived patterns, employing domain concepts to furnish an interpretative framework for human–machine interaction.

F. Data Processing

ML is adept at autonomously analyzing vast datasets to discern underlying patterns and leverage these findings to predict unseen data. In natural language processing, ML technologies assist in emulating human-like text processing capabilities. These technologies aid in extracting and amalgamating information from disparate datasets and sourcing and summarizing information from various origins. Breakthroughs in this domain

could revolutionize ML, significantly amplifying military intelligence processing efficacy. ML's proficiency in swiftly handling large volumes of data [39] to extract pertinent intelligence is invaluable for military personnel engaged in classification, pattern recognition, and correlation analyses [35].

G. Target Recognition

ML enhances target recognition accuracy within complex battlefield environments, enabling forces to garner comprehensive insights into prospective combat zones. Machine vision supplants the human eye for target recognition, tracking, measurement, and subsequent image refinement for analytical processing. Machine vision technology integrates a sequence of image processing operations with ML techniques to break down image analysis into more manageable subtasks. In addition, ML in target recognition systems advances the precision of identifying target coordinates and fosters intelligent research on battlefield spatial perception, concentrating on developing intelligent airborne (vehicle-borne, shipborne) systems with perceptive capabilities.

H. Battlefield Healthcare

The confluence of ML technology with emergent communication technologies like 5-G enables remote robotic surgery. AI applications extend to disease detection and genetic anomaly identification. They also offer analytical and therapeutic strategies for veterans suffering from post-traumatic stress disorder through AI-based psychotherapy analysis and treatment modalities.

The incorporation of human-machine interaction has facilitated more efficient and convenient application of acupuncture techniques such as, scalp acupuncture, auricular acupuncture, and abdominal acupuncture on the battlefield for pain relief, anti-inflammatory, antiallergic treatments, and in regulating soldiers' fighting spirit. This represents an innovative foray of AI into the realm of battlefield first aid. Furthermore, the integration of the holistic perspectives of Heaven-Man Unity and the holistic view of the human body with human-machine interaction technology has injected a novel impetus for original innovation in the development of AI.

I. Monitoring and Situational Awareness

Intelligence, surveillance, and reconnaissance (ISR) operations are pivotal for surveillance and situational awareness, supplying critical information for military maneuvers [35]. Autonomous systems executing ISR tasks can be remotely operated or navigate predefined routes. Endowing these systems with ML enhances warfighters' ability to monitor hazards and maintain situational awareness.

ML can intelligently extract time-sensitive intelligence from cyberspace, addressing the challenges of voluminous data and complexity faced by battlefield perception systems, thereby, enabling commanders to maintain real-time awareness of the battlefield environment.

J. Cybersecurity

Military systems are vulnerable to cyber threats that risk compromising confidential military information and impairing military systems. However, systems equipped with ML, such as network confrontation game AI algorithms and breakthrough AI systems, can autonomously safeguard networks, computers, programs, and data against unauthorized access [35]. Generative adversarial networks can produce multidimensional few-shot representations of cyberspace elements like logs, traffic, and executable programs.

VII. CONCLUSION

In the vortex of the current scientific and technological revolution, the industrial and informational revolutions have converged, catalyzing a novel paradigm in military intelligence. ML has emerged as a pivotal axis for nations to advance their military modernization agendas, igniting a global arms race in AI.

This article has delved into the intersection of ML with military applications and APTs within human-machine systems. As nations accelerate their strategic deployments, ML-driven forces are being rapidly integrated. Worldwide military infrastructures are assimilating ML technologies across terrestrial, maritime, aerial, and space-based platforms.

The Army, for instance, is pioneering distributed processing research and embedding ML systems within autonomous networks in diverse environments. The Navy is advancing autonomous naval information systems and crafting responses to electronic warfare initiatives. The Air Force's "Quantum Plan" and its investment in autonomous resilient networks, ML wingmen, and sixth-generation machinery are poised to substantially bolster combat capabilities. The Marine Corps is engaging in advanced reinforcement learning for collaborative information warfare, while the Space Force is enhancing space-based data management. In addition, socio-psychological human-machine systems have been meticulously examined.

This discourse has also spanned the IoBT, behavior detection via ML, social network analytics, and electronic warfare among other cyber threat dimensions. It has scrutinized the military applications of SDN, ICN, DTN, IoBT, and other technologies in human-computer interaction contexts. Furthermore, this article has collated the extensive potential of ML across critical domains such as, operations, intelligence, networking, logistics, recognition, medical aid, and situational awareness. In recent times, leading nations have engaged in AI research to foster innovation in combat theory and intelligent equipment development. This article also highlights AI preresearch initiatives undertaken in recent years.

Underpinned by AI-driven fusion, empowerment, knowledge reasoning, exploration, and interpretation, algorithms have demonstrated reliability. The Russia-Ukraine conflict has redefined the digital battlefield landscape, establishing a transparent battlefield environment, achieving intelligent cross-domain equipment synergy, and capturing the strategic high ground of modern combat.

This article has provided a lens to track and evaluate the evolution of global military ML technology. It offers a synthesis

of the current state of global military ML technology, dissects its principal technological trajectories, contemplates its developmental tendencies, investigates emblematic military application case studies, and projects the future landscape of global military ML technology.

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